

Information with SVGD

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1 Problem Definition and Notation

Assume we have an object defined by geometric parameters θ located at state x_t . Normally, a state consists of the a position q_t and a velocity v_t , but for simplicity our state will only consist of the position. The velocity can be imputed from backwards Euler finite differences: $v_{t+1} = q_{t+1} - q_t$, with the assumption $v_0 = 0$. We also have a robot that has taken trajectory r_t . Each timestep, we have also collected measurements m_t , and we have access to a normalized per-timestep measurement model $p_\theta(m_t|x_t)$. We also assume that we have access to a forward/backward state transition likelihood in the exponential family:

$$\log p_\theta(x_t|x_{t+1}) \propto f_\theta(x_t, x_{t+1}) \quad (1)$$

In practice, f is the violation-implicit (vimp) loss (without any measurement terms).

Assume we have previous run a learning algorithm to compute a maximum-likelihood estimate (MLE) of our learned parameters ω , which include the object state at the present timestep T : $\omega = (\theta, x_T)$. The maximum likelihood is marked with a tilde, e.g. $\tilde{\omega}$. During the learning process, we have also learned some maximum-likelihood trajectory $\tilde{x}_t$.

Our first objective is to compute the Observed Information $\mathcal{I}$, defined as the empirical variance of the gradient of the measurement likelihood evaluated at the MLE.

$$\mathcal{I}(\tilde{\omega}, m_t, r_t) := \frac{1}{T} \sum_t (\nabla_\omega \log p_\theta(m_t|x_T)) (\nabla_\omega \log p_\theta(m_t|x_T))^T \quad (2)$$

1.1 Measurement Model

Currently, each timestep there are two measurements per robot sensor $m_t = \{c_t, n_t\}$, the contact boolean $c_t \in \{0, 1\}$ (i.e. $c_t = 1$ if contact is made), and the contact normal $n_t \in SO(2)$ (though it is represented in R^3). We can directly write down the log measurement loss as follows:

$$\log p_\theta(m_t|x_t) = - \sum_{N_r} c_t \frac{1}{2} (\sigma_n^{-2} (1 - \hat{n}_t(\theta, x_t) \cdot n_t) + \log(2\pi\sigma_n^2)) \quad (3)$$

$$+ (c_t - 1) \alpha \phi_t(\theta, x_t) + \log(1 + e^{\alpha \phi_t(\theta, x_t)}) \quad (4)$$

Where:

- N_r is the number of robot sensors
- $\phi_t(\theta, x_t)$ is the signed distance between the object and each robot sensor
- $\hat{n}_t(\theta, x_t)$ is the calculated contact normal between the object and each robot sensor
- α and σ_n are variance hyperparameters

Notes:

- The constant drops out of the (3) for the gradient, but the direct value is needed for some *softmax* computations below.
- $-\log(\text{sigmoid}(-x)) = \log(1 + \exp(x))$. This is more numerically stable.

2 Algorithm

In Summary:

- Sample:** Given $\omega = (\theta, x_T)$, sample $x_{t < T} \sim p_\theta(\cdot|x_T)$ by recursively sampling from the (approximate) marginal distribution $\int p_\theta(x_t|x_{t+1})p_\theta(x_{t+1}|x_T)dx_{t+1}$.
- Gradient:** For each timestep, use the sampled states to compute the Jacobian of the measurement likelihood given the parameters: $\nabla_\omega \log p_\theta(m_t|x_T)$
- Return:** $\mathcal{I}$ as defined above

2.1 Sampling

Draw N_i samples of each timestep's state, with each sample index by i , such that $x_{t,i}$ is the i 'th sample at the t 'th timestep. We can do this through iterative Stein Variation Gradient Descent (SVGD), with s being the iteration number. We initialize from a Gaussian prior

with given covariance Σ_s about our maximum likelihood trajectory: $x_{t,i}^0 \sim \mathcal{N}(\tilde{x}_t, \Sigma_s)$. Then we iterate:

$$x_{t,i}^{s+1} = x_{t,i}^s + \frac{\epsilon}{N_i} \sum_j^{N_i} \left[k(x_{t,j}^s, x_{t,i}^s) \nabla_{x_{t,j}^s} \log p_\theta(x_{t,j}^s | x_T) + \nabla_{x_{t,j}^s} k(x_{t,j}^s, x_{t,i}^s) \right] \quad (5)$$

Where ϵ is the learning rate. The kernel k is defined as the radial basis function with the median bandwidth heuristic:

$$k(x, x') = \exp \left(-\frac{\|x - x'\|_2^2}{2h^2} \right) \quad (6)$$

Where $h = \tilde{d}^2 / \log N_i$, and $\tilde{d}$ is the median of the pairwise distances between all particles. Finally, the gradient log-likelihood $\nabla_{x_{t,i}^s} \log p_\theta(x_{t,i}^s | x_T)$ can be computed using the current samples of the following distribution, as follows:

$$\nabla_{x_{T-1}} \log p_\theta(x_{T-1} | x_T) = \nabla_{x_{T-1}} f_\theta(x_{T-1}, x_T) \quad (7)$$

$$\nabla_{x_t} \log p_\theta(x_t | x_T) \approx \nabla_{x_t} \log \sum_{x_{t+1,i} \sim p_\theta(x_{t+1} | x_T)} p_\theta(x_t | x_{t+1,i}) \quad (8)$$

$$= \nabla_{x_t} \log \sum_i \exp f_\theta(x_t, x_{t+1,i}) \quad (9)$$

$$= \text{softmax}_i(f_\theta(x_t, x_{t+1,i})) \cdot \nabla_{x_t} f_\theta(x_t, x_{t+1,i}) \quad (10)$$

Using the fact that $\nabla \log \sum \exp = \text{softmax}$, and the dot product happening over the i dimension.

2.2 Gradient Computation

Given per-timestep samples $x_{t,i}$, compute the Jacobian-measurement-log-likelihood $\nabla_\omega \log p_\theta(m_t | x_T)$. Importantly, we should ideally not take the gradient through the above sampling procedure (which can be unstable). Previous work has shown that the gradient can be written as follows:

$$\nabla_\omega \log p_\theta(m_t | x_T) = \left[\text{softmax}_i(\log p_\theta(m_t | x_{t,i})) - \frac{1}{N_i} \right] \cdot \nabla_\omega (\log p_\theta(m_t | x_{t,i}) + \log p_\theta(x_{t,i} | x_T)) \quad (11)$$

With the dot product happening over the i dimension.

Using this equation, we can also recursively define the state gradient-log-likelihood

$$\nabla_{\omega} \log p_{\theta}(x_{t,i}|x_T)$$

$$\nabla_{\omega} \log p_{\theta}(x_{T-1,i}|x_T) = \nabla_{\omega} f_{\theta}(x_{T-1,i}, x_T) \tag{12}$$

$$\nabla_{\omega} \log p_{\theta}(x_{t,i}|x_T) = \left[\text{softmax}_j(\log p_{\theta}(x_{t,i}|x_{t+1,j})) - \frac{1}{N_i} \right] \cdot \nabla_{\omega}(\log p_{\theta}(x_{t,i}|x_{t+1,j}) + \log p_{\theta}(x_{t+1,j}|x_T)) \tag{13}$$

$$= \left[\text{softmax}_j(f_{\theta}(x_{t,i}, x_{t+1,j})) - \frac{1}{N_i} \right] \cdot \nabla_{\omega}(f_{\theta}(x_{t,i}, x_{t+1,j}) + \log p_{\theta}(x_{t+1,j}|x_T)) \tag{14}$$

With the dot product happening over the j dimension.